

Event Tables



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Section Introduction-Event Tables Topics

- What is an Event table.
- Event Types supported by Event tables.
- Understanding `LOG_LEVEL` and `TRACE_LEVEL` when it comes to Events.
- LAB Create Event tables.
- Limitations of Event tables.
- Costs of Logging Events.
- Summary.



What is an Event Table ?

An Event table is a Snowflake table type which is designed to capture error messages or Information messages generated by Snowflake automatically or generated by a Snowflake provided logging functions used by a Snowflake Data Engineer in their code.(UDF, UDTF, Stored Proc)

Events can be captured when they are generated from UDF,UDTF or Stored Procedure.

Snowflake supports 2 EVENT types Log Events and Trace Events.

Log Events also called Log entries are generated by Snowflake provided logging functions used by a Snowflake Data Engineer in their code. You can create log entries to capture specific data situations like missing PK in a table that must have a PK or business scenario like low-inventory which can be detected by selecting data from the table.

Trace Events are generated and stored inside Event tables automatically by Snowflake to record the invocation of code like UDF, UDTF and Stored procedures.

Event logging must be enabled to be able to capture Log Events and Trace Events.

Event logging costs credits and storage and hence you need to enable it to be able to capture Log Events and Trace Events.

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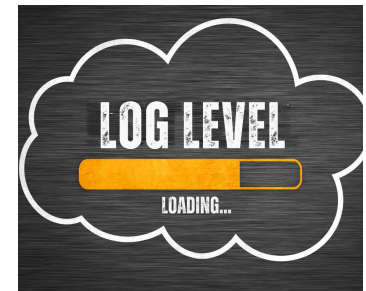
Understanding LOG_LEVEL.

Logging needs to be enabled to log Events, The value of LOG_LEVEL is set to OFF by default.

Logging is enabled by setting LOG_LEVEL to any value other than OFF Eg: SET LOG_LEVEL =INFO

LOG_LEVEL can take the following values when enabled.

- TRACE `SYSTEM$LOG('TRACE', ' Program has begun');`
- DEBUG `SYSTEM$LOG('DEBUG', ' In the IF Block');`
- INFO `SYSTEM$LOG('INFO','No Error, Employee PK is Fine ');`
- WARN `SYSTEM$LOG('WARN',Department_ID is nulll');`
- ERROR `SYSTEM$LOG('ERROR','Employee PK is NULL ');`
- FATAL `SYSTEM$LOG('FATAL', 'Employee table is empty ');`



Severity level of the message increases from TRACE to FATAL.

Snowflake Data Engineer code(UDF,UDTF and Stored Procedure) can use `SYSTEM$LOG` to log a message in the EVENT table.

If the LOG_LEVEL is set to INFO all SYSTEM\$LOG messages of higher severity will be logged in the EVENT table.

Understanding TRACE_LEVEL.

Trace Events are generated and stored inside Event tables automatically by Snowflake to record the invocation of code like UDF, UDTF and Stored procedures.

For Trace Events to be logged we need to setup the TRACE_LEVEL.
The value of TRACE_LEVEL is set to OFF by default.

TRACE EVENTS are enabled by setting TRACE_LEVEL to any value other than OFF

TRACE_LEVEL can take the following values when enabled .

- ON_EVENT.
- ALWAYS

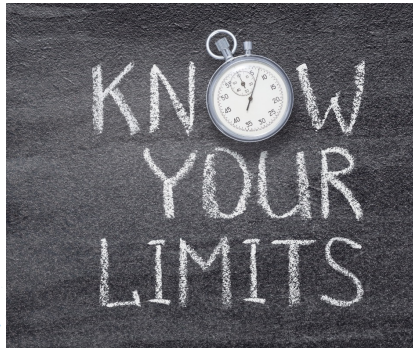
When TRACE_LEVEL =ON_EVENT, TRACE EVENTS are logged in the EVENT table if the Code (UDF,UDTF and Stored Procedure) generates LOG EVENTS([SYSTEM\\$LOG](#)) and adds them to the EVENTS table.

When TRACE_LEVEL =ALWAYS, TRACE EVENTS are logged in the EVENT table everytime the Code (UDF,UDTF and Stored Procedure) is called or invoked irrespective of if there are LOG EVENTS.



Limitation of EVENT Tables.

- EVENT tables can be used for logging only when being used from inside a UDF,UDTF or Stored Procedure.
- There is a lag of few mins before data from the LOG Event or Trace Event is available in the EVENT table.
- EVENT tables structure cannot be customized to add new columns to meet a specific use case for your business
- Adding an EVENT to a table does not result in any action or resolution of the challenges faced. The Events need to be analyzed and manual action could be needed in most cases to resolve issues.
- There is a 1 MB limit for LOG and TRACE event payloads. If the payload is over the 1MB threshold, the record in the event table will be incomplete.



Costs of EVENT Tables.

- Serverless credits to capture LOG and TRACE Events to the EVENT tables.
- To help reduce cost Snowflake collects the messages in batches and ingests the batches into the Event table.
- To reduce cost of logging avoid logging over a long period of time, enable logging only when needed.
- Logging can be done at different levels, you want to make sure that you log at the level needed so that you don't generate more logs than necessary.
 - Logging at the procedure level instead of database level will reduce logging ,events generated and cost.
 - Logging at session level to generate EVENTS only for the session rather than setting it permanently at the account, user, database, schema or procedure level will reduce logging ,events generated and cost.
- Monitor credit usage by going to the ACCOUNT_USAGE.EVENT_USAGE_HISTORY.
- Storage costs for data in EVENT tables will be charged to the account just like regular Snowflake tables. We should explore the possibility of removing data from the EVENT table after its useful life is complete.



Summary

- An Event table is a Snowflake table type which is designed to capture error messages or Information messages generated by Snowflake automatically or generated by Snowflake provided logging functions used by a Snowflake Data Engineer in their code.
- Events can be captured when they are generated from UDF,UDTF or Stored Procedure.
- Snowflake supports 2 EVENT types Log Events and Trace Events.
- Log Events also called Log entries are generated by a Snowflake provided logging function used by a Snowflake Data Engineer in their code.
- Trace Events are generated and stored inside Event tables automatically by Snowflake to record the invocation of code like UDF, UDTF and Stored procedures.
- Event logging must be enabled to be able to capture Log Events and Trace Events.
- The value of LOG_LEVEL is set to OFF by default. LOG_LEVEL can take the following values when enabled. TRACE, DEBUG, INFO, WARN, ERROR, FATAL.
- Severity level of the message increases from TRACE to FATAL.
- Snowflake Data Engineer code(UDF,UDTF and Stored Procedure) can use `SYSTEM$LOG` to log a message in the EVENT table.
- If the LOG_LEVEL is set to INFO all SYSTEM\$LOG messages of equal or higher severity will be logged in the EVENT Table.
- The value of TRACE_LEVEL is set to OFF by default. TRACE_LEVEL can take the following values when enabled ON_EVENT, ALWAYS.

Summary

- When TRACE_LEVEL is set to ON_EVENT, TRACE EVENTS are logged in the EVENT table if the code (UDF,UDTF and Stored Procedure) have LOG EVENTS in them([SYSTEM\\$LOG](#))
- When TRACE_LEVEL =ALWAYS, TRACE EVENTS are logged in the EVENT table everytime the code (UDF,UDTF and Stored Procedure) is called or invoked irrespective of if there are LOG EVENTS.
- There is a lag of few mins before data from the Log Event or Trace Event is available in the EVENT table, this is done to batch requests and save on credit consumption.
- EVENT tables structure cannot be customized to add new columns to meet a specific use case for your business.
- There is a 1 MB limit for LOG and TRACE event payloads. If the payload is over the 1MB threshold, the record in the event table will be incomplete.
- To reduce cost of logging avoid logging over a long period of time, enable logging only when needed. Log at level needed.
- Monitor credit usage by going to the ACCOUNT_USAGE.EVENT_USAGE_HISTORY.



Thank you



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